Task 10: Use Matplotlib module for plotting in python

Aim:

To use Matplotlib module for plotting in python

Problem: 10.1 Write a Python programming to display a bar chart of the popularity of programming langua

Sample Data:

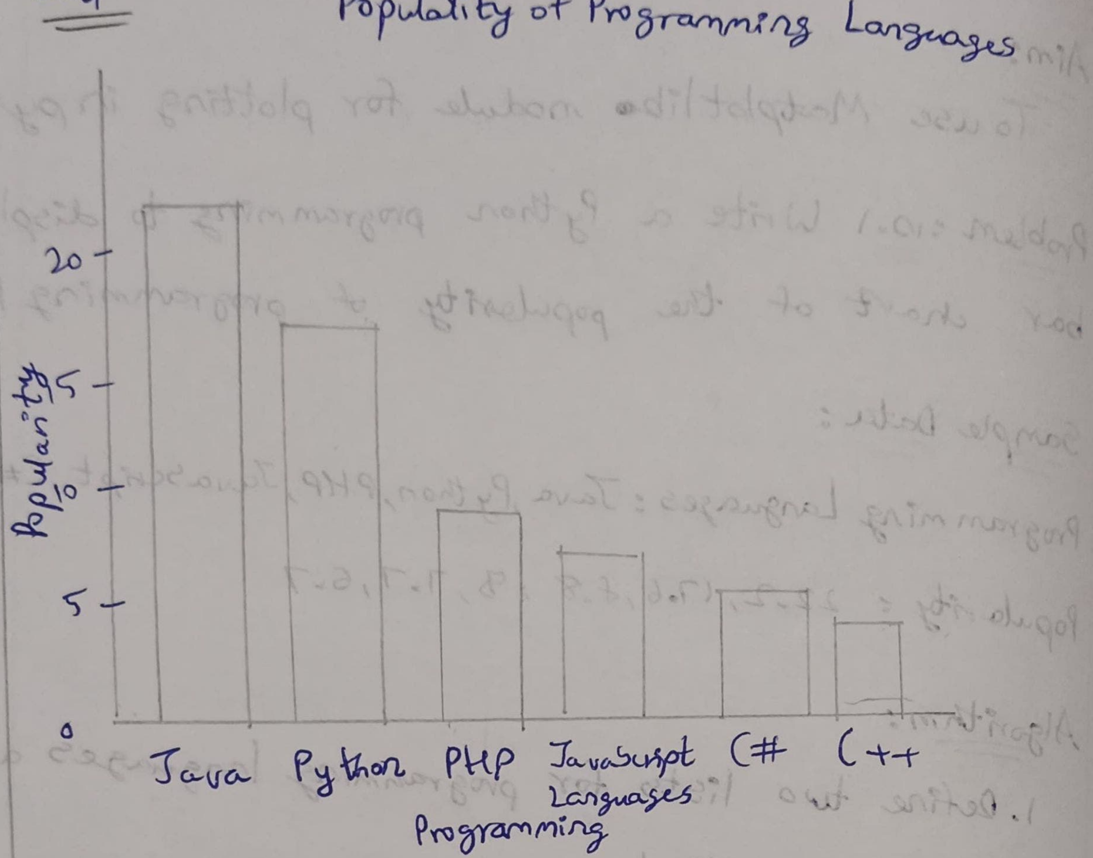
Programming Languages: Java, Python, PHP, Javascript, C#, C++

Popularity: 22.2, 17.6, 8.8, 8, 7.7, 6.7

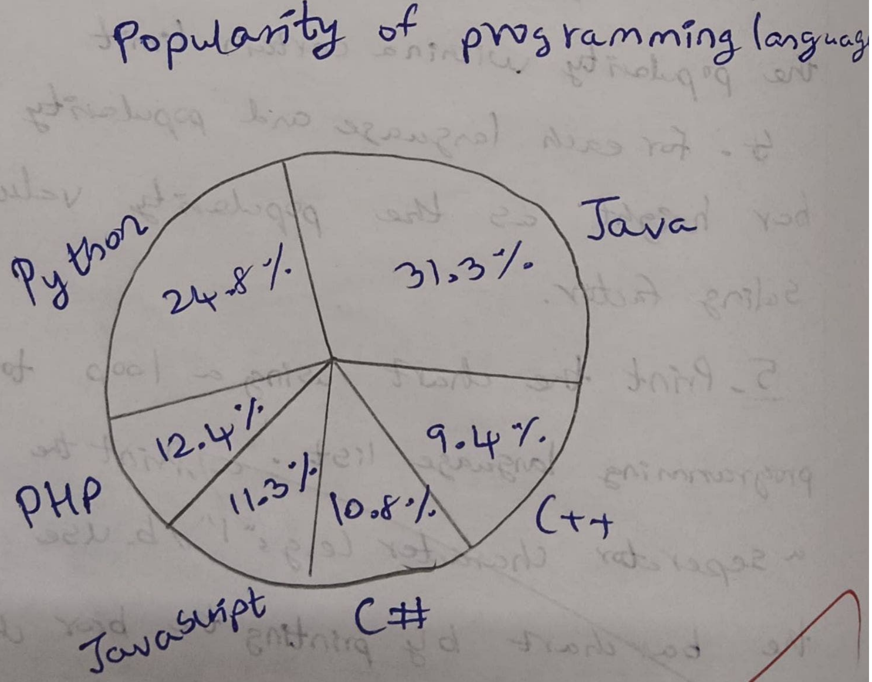
Algorithm:

1. Define two lists for programming languages and the popularity respectively
2. find the maximum popularity value in the list
3. Defining a scaling factor to scale the bar heights as the popularity within a certain limit
4. for each language and popularity pair, calulate the bar height as the popularity value scaled by the scaling factor.
5. Print the chart using a loop to iterate over the programming language list. a. Print the language name and a seperator character (e.g: "|") b. use a loop to print the bar chart by printing the bar character (eg: "x") a number of times equal to the bar height. c. Print the popularity value with a seperator character. d. Print a new line character

Output:



Output:



Program:

```
import matplotlib.pyplot as plt
languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]
plt.bar(languages, popularity, color='b')
plt.title("Popularity of Programming Languages")
plt.xlabel('Programming Languages')
plt.ylabel('Popularity')
plt.show()
```

Problem: 10.2

Write a python program to create a piechart of the popularity of programming languages.

Algorithm:

1. Create a list of programming languages & probebility.
2. Create a piechart using the matplotlib library
3. Set the title and beyond for the piechart
4. Show the piechart

Program:

```
import matplotlib.pyplot as plt
languages = ['java', 'python', 'PHP', 'Javascript', 'C#', 'C++']
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]
plt.pie(popularity, labels= languages autopt= '%.%.f%%')
plt.title('popularity of programming languages')
plt.legend(languages, loc='best')
plt.show()
```

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EX NO.	10
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	15

Result: Thus the python program using matplotlib